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Quantitative electroencephalography to assess post-stroke functional disability: A systematic review and meta-analysis

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ABSTRACT

Objective: Quantitative electroencephalography (QEEG) is a non-invasive, reliable and easily accessible modality to assess neuronal activity. QEEG in acute stroke may predict short and long-term functional outcomes. The role of individual indices has not been studied in a meta-analysis. We aim to assess individual QEEG-derived indices to predict post-stroke disability.

Methods: We included studies (sample size $\geq$ 10) with stroke patients who underwent EEG and a follow-up outcome assessment was available either in the form of modified Rankin scale (mRS) or National Institute of Stroke scale (NIHSS) or Fugl-Meyer scale (FMA). QEEG indices analysed were delta-alpha ratio (DAR), delta-theta-alpha-beta ratio (DTABR), brain symmetry index (BSI) and pairwise derived brain symmetry (pdBSI).

Results: Nine studies (8 had only ischemic stroke, and one had both ischemic and haemorrhagic stroke), including 482 participants were included for meta-analysis. Higher DAR was associated with worse mRS (n=300, Pearson's r 0.26, 95 % CI 0.21-0.31). Higher DTABR was associated with worse mRS (n=337, r=0.32, 95 % CI 0.26-0.39). Higher DAR was associated with higher NIHSS (n=161, r=0.42, 95 % CI0.24-0.6). Higher DTABR was associated with higher NIHSS (n=158, r=0.49, 95 % CI 0.31-0.67).

Conclusions: QEEG-derived indices DAR and DTABR have the potential to assess post-stroke disability. Adding QEEG to the clinical and imaging biomarkers in the acute phase may help in better prediction of post-stroke recovery.

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Introduction

Post-stroke motor paresis in the upper limbs is a common disability, and spontaneous recovery is expected in only 40 % of patients [1]. The current predictive models of motor recovery assist in prognostication and resource allocation [2,3]. The predictive models consist of patient factors (age, associated comorbidities, pre-existing cognitive impairment), clinical severity scores (baseline NIHSS, Motor examination including admission finger extension/Shoulder abduction, Fugl-Meyer assessment), imaging (infarct volume, small vessel disease) and neurophysiological assessment (motor evoked potential assessed with transcranial magnetic stimulation) [2]. The predictive accuracy is variable and probably needs a Bayesian approach with inputs from neurophysiological parameters [4]. Nijland et al[5] observed that voluntary finger extension and shoulder abduction within 72 h after stroke could give the probability of recovering some dexterity within six months, measured as scoring at least 10 points on the Action Research Arm Test. The prognostic value of Transcranial Magnetic Stimulation (TMS) for functional recovery in stroke was demonstrated by Di Lazzaro et al[6] who used the effect of theta burst stimulation on corticomotor excitability to predict

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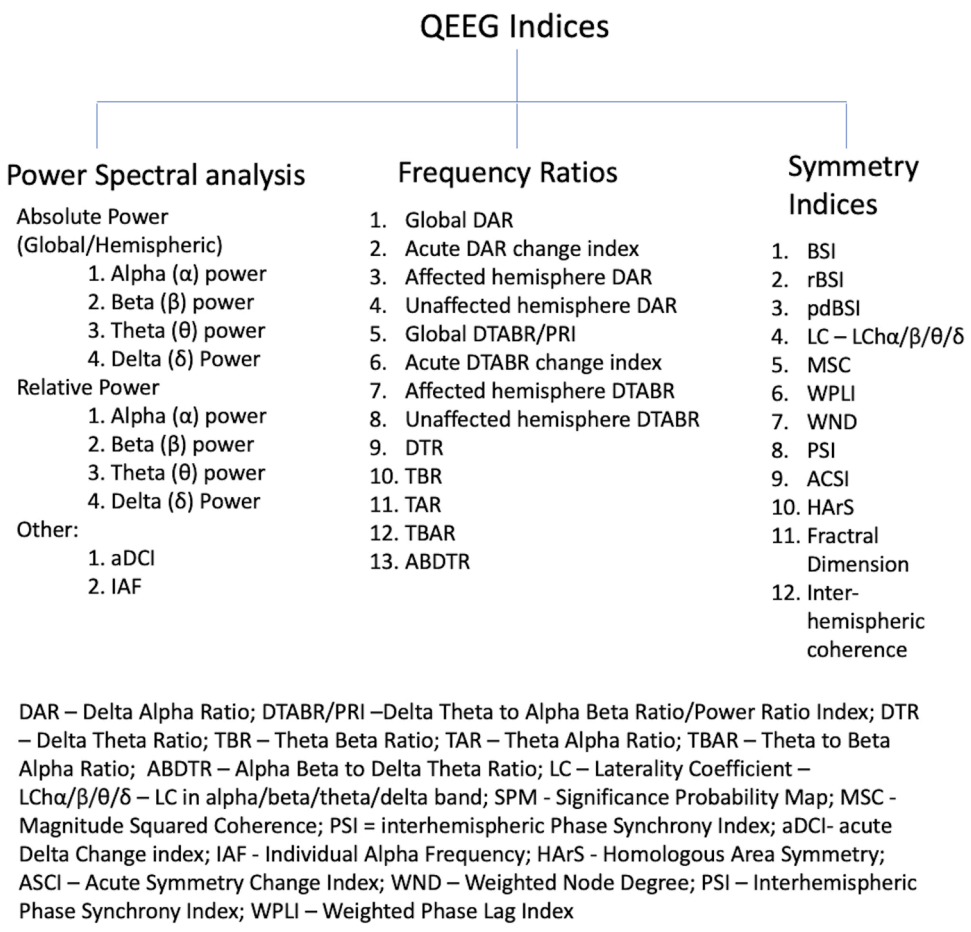


Fig. 1. Classification of QEEG indices

complete (mRS = 0) or partial (mRS > 0) recovery after stroke. The prevailing scoring systems for predicting prognosis after stroke are detailed in a review. [7]. The review suggest the Predict Recovery Potential (PREP) 2 score as the practical tool amongst the three externally validated scoring systems (PREP2, PRESS (Predictive Swallowing Score), GRAVo (Glasgow Coma Scale, Race, Age, Hematoma Volume). The PREP2 algorithm is a decision tree model that predicts recovery in stroke patients. It uses TMS and hence cannot be applied in clinical settings without this modality [7]. TMS studies are typically performed in specialized electrophysiology laboratories with the help of trained personnel. There is a need for non-invasive, easily accessible, and reliable adjunct tools to provide clinically meaningful motor outcome prediction at different time points [7,8]. The potential alternatives to TMS for bedside neurophysiological assessment include near-infrared spectroscopy (NIRS), electroencephalography (EEG), and evoked potential studies. NIRS has a low temporal resolution and fewer channels available. Evoked potential studies also have similar limitations in terms of channels. EEG has a high temporal resolution and more brain coverage. It is readily available at most sites with trained personnel.

Electroencephalography (EEG) is an affordable, high-resolution and real-time investigation of cerebral activity. A reactive cerebral cortex (normal) is more likely to express alpha and beta frequency, and a less active cerebral cortex (e.g., ischemic tissue) is more likely to express theta and delta frequencies depending on the volume of affected tissue [9,10]. Quantitative EEG (QEEG) measures allow the reduction of the inter-observer variability prevalent in qualitative EEG assessment [11]. The frequency analysis sub-technique of QEEG involves measuring the power of each major frequency band (i.e., alpha, beta, theta and delta), which is proportional to the amplitude/voltage produced in that band (Fig. 1) [12]. This can be computed as absolute power i.e., corresponding to the amplitude of each individual band or as relative power i.e., the amplitude of one frequency band is divided by the total voltage/power of all frequency bands [12]. Absolute power is preferred over relative power indices as they are a percentage of the whole [13]. Global absolute delta band power is indirectly correlated with good outcomes [14,15] and global absolute alpha power is directly correlated with good outcomes [16,17]. These frequency measures can further be used to calculate quantitative indices like Delta Alpha Ratio (DAR), Delta Theta to Alpha Beta Ratio (DTABR), Alpha Beta to Delta Theta Ratio (ABDTR) at each electrode channel, or can be averaged across all channels. Variations of these indices can be global (both hemispheres) or hemispheric (affected and unaffected). The brain symmetry Index (BSI), pairwise derived BSI (pdBSI), laterality coefficient, and magnitude squared coherence reflect asymmetry between cerebral hemispheres. Asymmetry can occur due to underlying remote and/or new stroke-related cerebral dysfunction. BSI values vary from 0 (complete symmetry) to 1 (complete asymmetry). More than >30 QEEG indices have been described and validated in studies (Appendix 1) (Fig. 1). Any change in one frequency waveform will affect the interpretation of other waveforms. The discussion about individual QEEG indices is beyond the scope of this study however in Appendix 1 we have highlighted the indices and derivation methods.

The prognostic role of qualitative or quantitative EEG indices in various stroke outcomes has been studied previously. A scoping review by Sutcliffe et al [9] noted surface EEG's diagnostic and prognostic utility in acute stroke. It can help predict dependency, later neurological impairment, cognitive function and mortality. A systematic review done by Doerffus et al [18] lists whether qualitative or quantitative EEG was performed in the study and the inference drawn with respect to functional outcome, mortality, post-stroke cognitive decline and post-stroke epilepsy. Lanzone et al [19] in a systematic review enlisted QEEG features

consistently related to stroke recovery, including clinical scores, aphasia and cognitive outcome. Vatinno et al [20] in a meta-analysis further used QEEG as a global measure to predict post-stroke disability. However, it would be pragmatic to understand the prediction capability of individual QEEG indices for practical utility. Thus, we aim to assess the role of individual QEEG-derived indices to predict post-stroke disability.

Methods

The Preferred Reporting Items for Systematic Reviews and Meta-analysis (PRISMA) guidelines for systematic reviews and meta-analysis [21,22] were followed. The study was registered in the International Prospective Register of Systematic Reviews (PROSPERO 2022 CRD 42022292281).

Search strategy

The databases used for the literature search were PubMed (Medline), Scopus, OVID, CINAHL databases and an additional source was from a previously published meta-analysis *(Supplementary File: Table 1)*. The search terms used were stroke and QEEG, and their related synonyms and abbreviations to keep the search strategy sensitive. Papers included in the search were from inception until October 07, 2022.

Eligibility criteria

Studies were included if they were original articles in English from peer-reviewed journals; papers assessing the relationship between EEG and subsequent stroke-related motor or functional outcomes; papers that reported statistical analysis results for prognosis and/or provided data sufficient for independent statistical analysis for prognosis; patients studied were both males or females presenting with ischemic stroke or intracerebral haemorrhage; there should be a minimum 10 patients included in the study. Studies were excluded if they did not report the association between EEG and clinical outcomes measures. Papers with meta-analyses, literature reviews, clinical guidelines, case studies, and trial protocols were excluded. Studies with patients that had a transient ischemic attack or subarachnoid haemorrhage were excluded. Stroke were classified as acute (within 7 days), subacute (7 days to 6 months), and chronic (>6 months).

Data extraction

Preliminary screening was done by three reviewers and papers were selected based on the title and abstract. Following the preliminary screening, the abstracts and titles were screened, and the duplicates were removed. The full text for eligible papers was obtained and screened independently by two reviewers. Papers were excluded for lack of follow-up outcome measures of interest, and papers that were not in English (Fig. 1). The papers that fulfilled the criteria were then included in the systematic review. Study characteristics were extracted from the selected papers, including patient characteristics, time since stroke, medical treatment, EEG protocol, EEG indices, functional outcomes type and time.

Risk of bias assessment

The quality assessment of the included studies was conducted based using the Quality in Prognosis Studies (QUIPS) tool [23,24]. An independent assessor assessed the studies. They were scored as low risk of bias, high risk of bias, or medium risk of bias in the domains of study participation, study attrition, prognostic factor measurement, outcomes measurement, study funding, statistical analysis, and reporting. Studies that had clear descriptions of the population (onset to enrolment, age, sex), setting (type of stroke, severity of stroke), and measures (EEG analyses, EEG indices); sufficient description of how the measure was executed and instrumentation used (EEG channels description, number of EEG performed); interpretation of the measure (availability of EEG indices median or mean values or only statistical analysis with beta values for linear regression or Student's t test or correlation analysis); use of specific threshold values (receiver operator curve analysis available); and the evaluation of the risk of serious injuries were considered to have minimal risk of bias and were rated as low risk of bias [24]. Studies that did not meet all the criteria of the low risk of bias but with no major bias and missing information, thus making it difficult to assess the limitations and potential problems, were considered to have a medium risk of bias [24]. In studies with significant flaws and biases in the design, analysis, or reporting, large amounts of missing data and reporting discrepancies were considered to have a high risk of bias. Studies with a high risk of bias were not excluded but were considered to be less reliable than studies with a low or medium risk of bias [24]

Data analyses

The most common measure of effect size in abstract studies for metanalysis was the correlation coefficient. For studies that didn't have a correlation coefficient, the odds ratio was used to compute the correlation coefficient. Studies that had multiple EEG acquisitions at different time points were analysed separately. Studies that had multiple outcome assessment at different time points were analysed separately. The random effects model was used since we expected heterogeneity to be present due to known differences in stroke patient characteristics and variable follow-up duration. The standard error and Fisher's Z value were computed. Meta-analyses for QEEG indices (DAR, DTABR) were conducted if at least 3 independent QEEG observations (at baseline and at follow-up) were correlated with the functional outcome measure (modified Rankin scale, mRS; National Institute of Health Stroke scale, NIHSS; Fugl Meyer assessment, FMA). The overall effect estimate, 95 % CI of the effect size, weight, weight % per QEEG index were used to create summary forest plots for functional outcome measures. (IBM Corp. Released 2021. IBM SPSS Statistics for Mac, Version 28.0. Armonk, NY: IBM Corp)

Results

Systematic review syntheses

A total of 2143 papers were identified from 5 databases, Pubmed, Ovid, CINAHL/EBSCO, and Scopus (Fig. 2). 29 papers fulfilled the inclusion criteria and were included in the systematic review [14–17,25–50]. The summary of the characteristics of these studies were tabulated according to the following main clinical outcomes of interest; mRS (Table 1) NIHSS (Table 2) and FMA (Table 3). The studies included in the systematic review were assessed for eligibility for meta-analysis. There were 9 studies that fulfilled the inclusion for the meta-analysis and were subsequently analysed [14,16,17,26,30,31,38,40,49]. In cases of discrepancy, resolution was found by a consultation. QEEG indices are reported in the pattern of power spectral analysis, frequency ratios and symmetry indices.

A total of 14 studies used mRs as the primary outcome assessment method [14,16,26,31,32,34–38,40,42,49,50]. All studies had patients with first-ever acute ischemic stroke except Rogers et al which also included patients with intracerebral haemorrhage (ICH). EEG acquisition was done ≤4.5 h (n=1), ≤ 24 h (n=2), ≤48 h (n=3), ≤72 h (n=7), ≤7 days (n=1) and ≤10 days (n=1). EEG acquisition was done with help of a single channel (n=1), 16 channels (n=5), 19 channels (n=4), 21 channels (n=1) and 64 channels (n=2) machine. mRS was assessed at time points ranging from 7 days (n=1) to 12 months (n=3). mRS was analysed as a dichotomous (n=6) or ordinal variable (n=8). Relative theta Power (TP) and Delta Power (DP) were directly correlated with mRS[53,54]. DAR, interhemispheric DAR difference, and DTABR were directly correlated and ABDTR was inversely correlated with mRS [53,55,56]. Pairwise derived brain symmetry index (pdBSI), and Brain

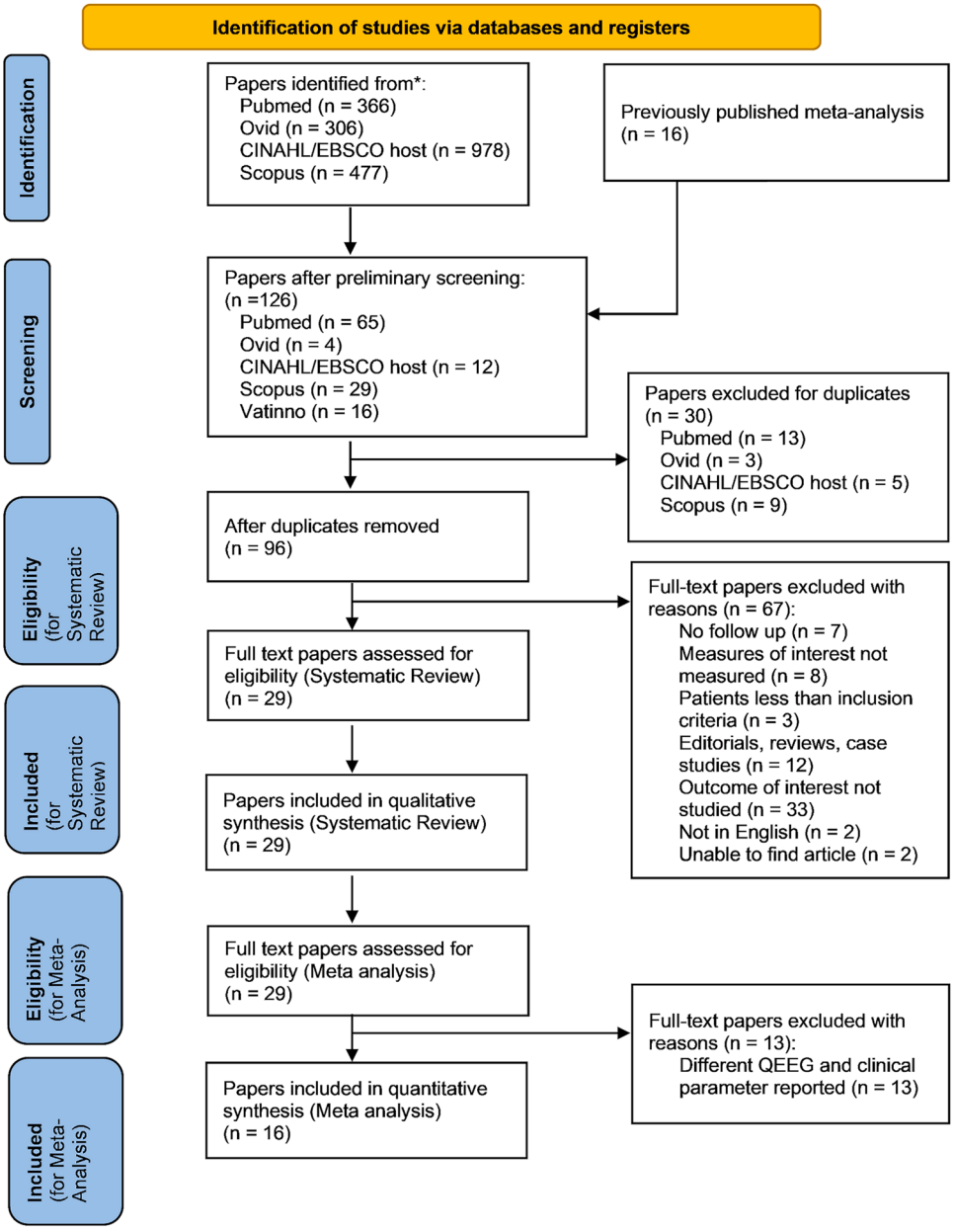


Fig. 2. Preferred Reporting Items for Systematic Reviews and Meta-analysis (PRISMA) 2020 flow diagram for new systematic reviews which included searches of databases and registers only. CINAHL, Cumulative Index to Nursing and Allied Health Literature; EBSCO Elton B Stephens Company; QEEG, Quantitative Electroencephalography

Table 1
Characteristics of studies acquiring quantitative electroencephalograph and assessing modified Rankin scale as an outcome measure

Author, Year, (Sample Size)	Stroke Type, enrolment	Quantitative EEG Indices (EEG Channels)	Clinical Outcome		Results
			Tool	Outcome Assessment time	
Dickey et al (2022) (n=73)	I (LVO), acute < 72 hrs	DAR (16)	mRS, NIHSS	3 months	Absolute Interhemispheric DAR difference directly correlated with mRS & NIHSS.
Zhang et al (2021) (n=29)	I, acute before MT and 12h, 24h and 7 days after MT	AP, DAR, DTABR (16)	mRS≤ 2, NIHSS	3 month mRS, 7 day NIHSS	DP, DAR, DTABR directly correlated with mRS. DP, DAR, DTABR before MT, 24h and 7days after MT directly correlated with NIHSS.
Ajcevic et al (2021) (n=20)	I, within 4.5h	PSD, AP, RP, DAR, DTABR (64)	mRS≤2, NIHSS	mRS at12 months, NIHSS at 7 days	DAR is associated with mRS. DP, TP, DAR, DTABR directly correlated with NIHSS. AP, BP inversely correlated with NIHSS.
Tian et al (2021) (n=30)	LHI, acute < 72 hrs	PSD, AP, ABDTR index (19)	mRS	3 months	ABDTR inversely correlated with mRS.
Wang et al., 2021 (n=31)	I, <48 h after EVT	AP, RP, DAR, DTABR (?)	mRS≤ 2	3 months	Higher DAR and DTABR are associated with poor outcome. DAR is a better predictor of outcome than DTABR.
Jeffrey Rogers et al., 2020 (n=16)	I+H, <72h	AP, RP, DAR, DTR, DTABR (1)	mRS, mBI, NIHSS	1 & 3mths	RP_{theta} directly correlated with 30 and 90 day mRS, mBI.
Bentes et al., 2018 (n=151)	I, < 72 hrs	RP, DTABR, DTABR (aff & unaff), DAR, ACSI, Acute DTABR change index (64)	mRS≥3	12 months	DTABR and Alpha RP have a higher discriminative capacity to predict outcomes. Higher DTABR is 1.7 times more associated with poor outcome
Van Kaam et al., 2018 (n=18)	I, <48 h	MSC, WPLI, DAR (21)	mRS≥3	3 months	Contralateral WPLI and MSC in the alpha band in poor outcome group were lower than those in the good outcome group.
Xin et al., 2017a (n=29)	I, < 72 hrs	rBSI (16)	mRS≤2 and BI>60	21 days	Higher rBSI associated with poor outcome.
Xin et al., 2012 (n=22)	I, <48 h	BSI (16)	mRS	1 month	BSI directly correlated with mRS.
Sheorajpanday et al., 2011a (n=110)	I, <24 h	DTABR, pdBSI (19)	mRS	6 months	DTABR, pdBSI directly correlated with mRS.
Sheorajpanday et al., 2011b (n=22)	I, LACS < 72 hrs	DTABR, pdBSI (19)	mRS	Day 7	DTABR directly correlated with mRS.
Cuspineda et al., 2007 (n=28)	I, < 72 hrs	AE (19)	mRS	3 months	Delta AE is best variable for predicting long term outcome.
Cillessen et al., 1994 (n=55)	I, 1-10 days	EEG peak (16)	mRS	12 months	EEG correctly predicted a good outcome in 39 of 41 patients and a poor outcome in 11 of 14 patients

Footnotes: Abbreviations: AP- Absolute Power, RP – Relative Power, DAR – Delta Alpha Ratio, DTR – Delta Theta Ratio, DTABR – Delta Theta to Alpha Beta Ratio, mRS – Modified Rankin Scale, mBI – Modified Barthel Index, NIHSS – National Institute of Health Stroke Scale, ASCI – Acute Symmetry Change Index, pdBSI – pairwaise derived Brain Symmetry Index, PSD - Power Spectral Density, ABDTR – Alpha Beta to Delta Theta Ratio, MSC – Magnitude Squared Coherence, WPLI – Weighted Phase Lag Index, BSI - Brain Symmetry Index, rBSI – revised Brain Symmetry Index, DP – Delta Power, MT – Mechanical Thrombectomy

Table 2
Characteristics of studies acquiring quantitative electroencephalograph and assessing National Institute of Health stroke scale as an outcome measure

Author, Year, (Sample Size)	Stroke Type, enrolment	Quantitative EEG Indices (Channels)	Clinical Outcome		Results
			Tool	Outcome Assessment time	
Zappasodi et al., 2019 (n=120)	I, within 10 days	PSD, IAF, HArS (19)	NIHSS, ER	6 months	Smaller DeltaTheta interhemispheric symmetry in acute phase correlates with a better clinical picture in stabilized phase. Higher C/L DeltaTheta power correlated with worse clinical recovery.
Nicolo et al., 2015 (n=24)	I+H, acute	WND (128)	NIHSS	3 months	I/L and C/L WND is an independent predictor of motor improvement
Zappasodi et al., 2014a (n=36)	I, between 4 – 10 days	Fractal dimension, PSD (19)	NIHSS	6 months	FD of lesioned hemisphere inversely correlated with NIHSS
Assenza et al., 2013a (n=42)	I, between 4 – 10 days	PSD, Interhemispheric coherence (19)	NIHSS, ER	6 months	I/L, C/L DP and TP directly correlated with NIHSS. I/L, DP and TP inversely correlated with ER. C/L AP, BP, DP, TP and whole spectrum power correlated negatively with ER
Sheorajpanday et al., 2011b (n=29)	I (POCS), within 72h	DTABR, pdBSI (19)	NIHSS	Day 7	pdBSI directly correlated with NIHSS.
Finnigan et al., 2007a (n=13)	I, 49±3h post symptom onset	AP, DAR, PRI (64)	NIHSS	30 days	DAR, PRI directly correlated with NIHSS. RAP inversely correlated with NIHSS.
Finnigan et al., 2004b (n=11)	I, within 2h after 1st MRI and just before 2nd MRI	AP, aDCI (64)	NIHSS	30 days	aDCI directly correlated with NIHSS

Abbreviations: DTABR – Delta Theta to Alpha Beta Ratio, pdBSI – pairwise derived Brain Symmetry Index, NIHSS – National Institute of Health Stroke Scale, AP – Absolute Power, aDCI – acute Delta Change Index, DAR – Delta Alpha Ratio, PRI – Power Ratio Index, RAP – Relative Alpha power, PSD – Power Spectral Density, ER – Effective recovery, I/L – Ipsilesional network interaction, C/L – Contralesional network interaction, DP – Delta Power, TP – Theta Power, BP – Beta Power, WND – Weighted Node Degree, IAF – Individual Alpha Frequency, HArS – Homologous Area Symmetry

Table 3
Characteristics of studies acquiring quantitative electroencephalograph and assessing Fugl-Meyer assessment as outcome measure

Author, Year, (Sample Size)	Stroke Type, enrolment	Quantitative EEG (Indices)	Clinical Outcome		Results
			Type	Outcome Assessment time	
Saes et al., 2021 (n = 39)	I, within 3 weeks	DAR, BSI (62)	FMA, NIHSS	26 weeks	Higher BSItheta predicts lower FMA.
Sebastián-Romagosa et al., 2020[51] (n=34)	I+H, Chronic	BSI, LC (16)	FMA, BBT, HPT, BI	1 month	BSI inversely correlated with FMA. LChα, LChβ inversely correlated with FMA.
Kawano et al., 2020 (n=40)	I, >2 weeks of onset	PSI (19)	FMA, NIHSS, FIM	30 days (FM-UE1), 120 days (FM-UE2)	C3C4-PSI directly correlated with FM-UE1 in θ, α, β1, β2 bands. Combination of CST-LV and C3C4-PSI predicted FM-UE1 better than either alone. P3P4-PSI directly correlated with FM-UE1 in α band. Contralesional IntrahC-PSI directly correlated with FM-UE gain in δ and θ bands.
Mane et al., 2018 - tDCS BCI group[52] (n=10)	I+H, Chronic (9 months)	PSD, AP, RP, DAR, TBR, TAR, TBAR, BSI, pdBSI (27)	FMA, ΔFMA	2,4 weeks	RDP, PRI, DAR inversely correlated with ΔFMA (2wks) in resting state. PRI indirectly correlated with ΔFMA (2wks) in task state. RDP, PRI, DAR inversely correlated with ΔFMA (4wks) in resting and task state.
Guggisberg et al., 2017 (n=63)	I, acute	WND (128)	FMA	3 months	Beta-band WND>0.1 at the I/L ventral premotor area is showed PROP in 80 % patients. Beta-band WND<0.1 at the I/L ventral premotor area showed POOR in 78 % patients.
Trujillo et al., 2017a (n=10)	I+H, Chronic	PSD, AP, RP, DAR, PRI, pdBSI (64)	FMA, ΔFMA, ΔFMA %	4 weeks	PRI inversely correlated with ΔFMA and ΔFMA %. RDP, RTP inversely correlated with ΔFMA. RAP directly correlated with ΔFMA.
Agius Anastasi et al., 2017a (n=12)	I, subacute	BSI (32)	FMA, MI	variable for each patient - max duration 4 months for 3 patients	BSI inversely correlated with FMA (2months) and MI (1month).
Ang et al., 2015a (n=11)	I+H, Chronic after 3 months	rBSI (27)	FMA	4 weeks	rBSI inversely correlated with FMA

Abbreviations: BSI – Brain Symmetry Index, LC – Laterality Coefficient, LChα – Laterality Coefficient of healthy hand in alpha band, LChβ - Laterality Coefficient of healthy hand in beta band, PSD – Power Spectral Density, AP – Absolute Power, RP – Relative Power, DAR – Delta Alpha Ratio, TBR – Theta Beta Ratio, TAR – Theta Alpha Ratio, TBAR – Theta to Beta Alpha Ratio, pdBSI – pairwise derived Brain Symmetry Index, DTABR – Delta Theta to Alpha Beta Ratio, PRI – Power Ratio Index, WND – Weighted Node Degree, PSI – Phase Synchrony Index, FMA – Fugl Meyer Assessment, BBT – Box and Block Test, HPT – 9 Hole Peg Test, BI – Barthel Index, NIHSS – National Institute of Health Stroke Scale, MI – Motricity Index, FIM – Functional Independence Measure, RDP – Relative Delta Power, PRI – Power Ratio Index, RTP – Relative Theta Power, RAP – Relative Alpha Power, BSItheta – Brain Symmetry Index, CST-LV – Corticospinal tract – Lesion volume, P3P4-PSI – Parietal – Parietal Phase Synchrony Index, IntrahC-PSI – Intrahemispheric Phase Synchrony Index centered on C3/C4

symmetry index (BSI) were directly correlated with mRS [57,58].

A total of 7 studies used NIHSS as the main outcome assessment method [15,17,43,44,48,49]. All studies had patients with first-ever acute ischemic stroke except Nicolo et al which also included patients with ICH. EEG acquisition was done ≤ 4.5 h (n=1), ≤ 48 h (n=1), ≤72 h (n=3), ≤ 7 days (n=1) and ≤ 10 days (n=3). EEG acquisition was done with help of a single channel (n=1), 16 channels (n=2), 19 channels (n=4), 64 channels (n=3) and 128 channels (n=1) machine. NIHSS was assessed at time points ranging from 7 days (n=2) to 12 months (n=1). Ipsilateral, Contralateral, and global DP and TP were directly correlated with NIHSS[59]. The acute delta change index (aDCI) was directly correlated with NIHSS. Alpha power (AP), Relative AP and Beta Power were inversely correlated with NIHSS (Assenza et al., 2013b; Finnigan et al., 2004c). DAR, DTABR and absolute interhemispheric DAR differences were directly correlated with NIHSS [59,60]. The pdBSI was directly correlated and the fractal dimension size of the lesioned hemisphere was inversely correlated with NIHSS [58,61].

A total of 8 studies used FMA as the main outcome assessment tool [25,28–30,33,39,45,47]. These included acute (n=1), subacute (n=3) and chronic (n=4) stroke patients. Four studies had ischemic stroke patients and four studies had patients with ischemic stroke and ICH. EEG acquisition was done with help of 16 channels (n=1), 19 channels (n=1), 27 channels (n=2), 32 channels (n=1), 62 channels (n=1), 64 channels (n=1) and 128 channel (n=1). FMA was assessed at time points ranging from 2 weeks (n=1) to 6 months (n=1). Relative DP, power ratio index, DAR, BSI, and relative BSI are inversely correlated with follow up FMA [62–64]. Relative Alpha Power (AP) is directly correlated with change in FMA [65].

Risk of bias assessment of included studies

About half of the studies had a source of bias (Fig. 3). Thus, half of the studies with low or relatively low risk of bias were used to perform subgroup analysis. All the studies reported the prognostic measures and outcome measures well enough. One study did not describe the study population adequately and study attrition[29], while another two study did not adequately describe the study confounders [25,33]. The remaining 6 studies did not adequately describe the study populations or the attrition or the statistical analysis did not report strategies for model building [17,30,31,33,36,40].

Meta-analyses

A total of 9 studies with a sample size of 482 were included in the meta-analyses. In 5 studies (n=300) we explored the correlation between early DAR in patients with stroke to follow-up mRS (90 days, n=4 or 12 months, n=1). Global DAR (n=5) was the most common index, followed by affected hemisphere DAR (n=1), unaffected hemisphere DAR (n=1) and interhemispheric DAR (n=1). DAR was acquired earliest ≤ 24h and latest ≤7 days. The DAR was directly correlated to follow-up mRS (Pearson's r = 0.29, 95 % CI 0.21-0.36) (Fig. 4A). In 5 studies (n=337) we explored the correlation between early DTABR in patients with stroke to follow-up mRS (90 days, n=3 or 6-month, n=1 or 12 months, n=1). Global DTABR (n=7) was the most common index, followed by affected hemisphere DTABR (n=1) and unaffected hemisphere DTABR (n=1). DTABR was acquired earliest ≤24 h and latest ≤7 days. The DTABR was directly correlated to follow-up mRS (r = 0.35, 95 % CI 0.24-0.46) (Fig. 4B).

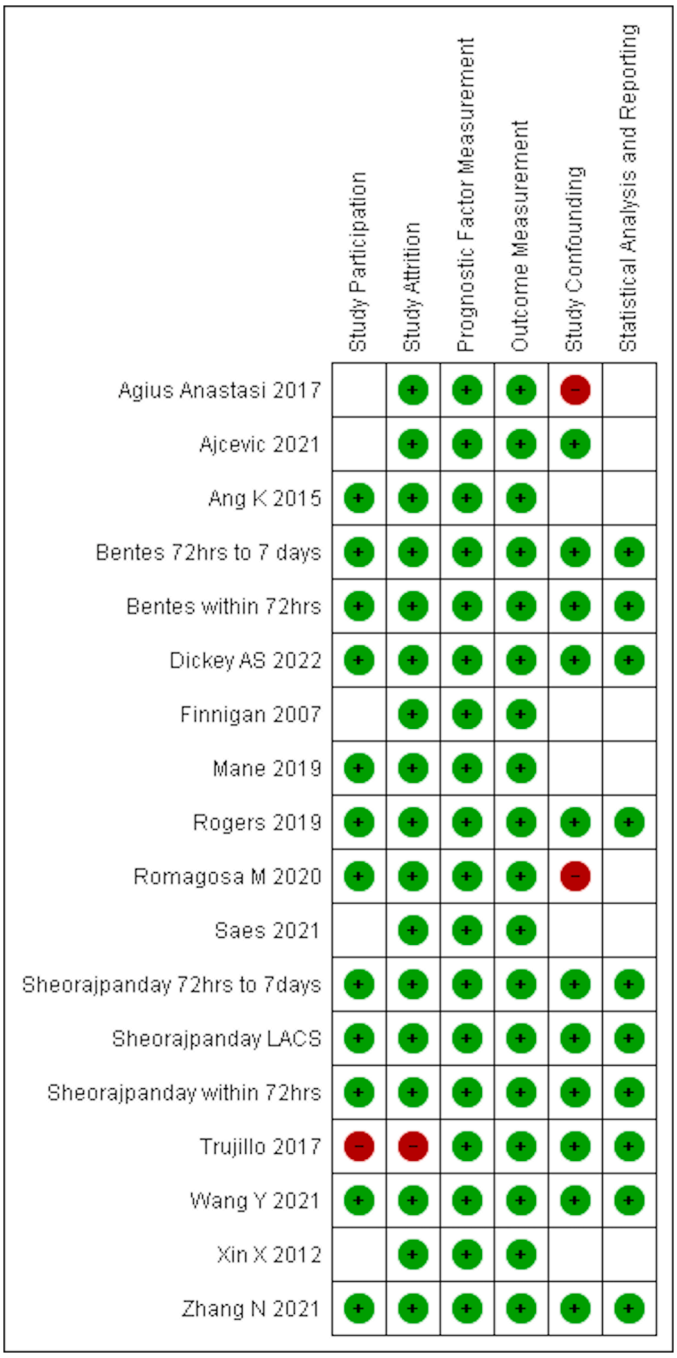


Fig. 3. Risk of Bias summary. LACS, Lacunar Stroke Syndrome.

In 4 studies (n=161) we explored the correlation between early global DAR in patients with stroke to follow-up NIHSS (7 days, n=5 or 90 days, n=1 or 6 months, n=1 or 12 months, n=1). DAR was acquired earliest ≤ 4.5h and latest ≤ 21 days. The DAR was directly correlated to follow-up NIHSS (r 0.54, 95 % CI 0.32-0.75) (Fig. 5A). In 4 studies (n=158) we explored the correlation between early global DTABR in patients with stroke to follow-up NIHSS (7 days, n=3 or 30 days, n=1 or 6 months, n=1 or 12 months, n=1). DTABR was acquired earliest ≤ 4.5 h and latest ≤ 72 h. The DTABR was directly correlated to follow-up NIHSS (r = 0.52, 95 % CI 0.32-0.71) (Fig. 5B). A meta-analysis exploring the correlation between FMA and QEEG indices could not performed as no minimum three studies were satisfying the eligibility criteria.

Discussion

In this systematic review and meta-analysis, we noted a modest correlation between QEEG indices with follow-up functional outcome measures, which included power spectral analysis indices (DP, TP were directly correlated with worse outcomes and AP, Beta Power were inversely correlated with worse outcomes), frequency ratio indices (DAR, DTABR were directly correlated with worse outcomes) and symmetry indices (BSI, pdBSI were directly correlated with worse outcomes). In the meta-analysis, early DAR and DTABR had a weak-positive correlation with mRS and a moderate-positive correlation with NIHSS in patients with acute stroke.

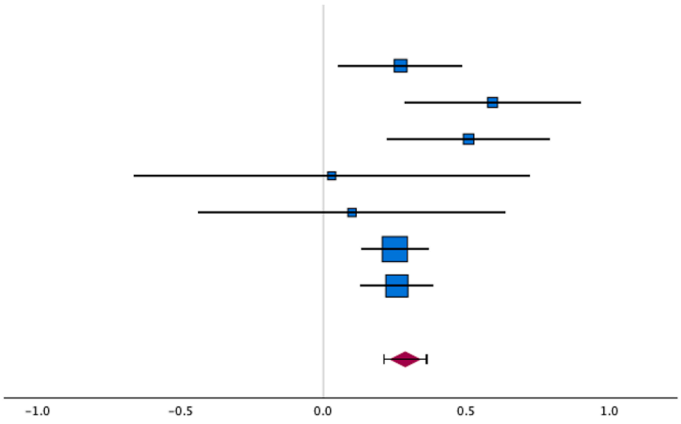
In a study by Finnigan et al. of 28 control subjects and 18 acute ischemic stroke patients (EEG acquired <24 h of symptom onset), global DAR in control subjects was 1.3±1 and in patients, it was 6.6 ± 2.8 [66]. Global DAR >3.7 had 100 % sensitivity and 100 % specificity to discriminate between ischemic stroke patients and control subjects. In the same study, DTABR in control subjects was 0.8 ± 0.5 and in patients was 4.3 ± 1.8. DTABR >1.8 had 100 % sensitivity and 96 % specificity. In a study done by Bentes et al. (n = 151), EEG was acquired <72 h of ischemic stroke symptom onset, and mRS was assessed at 12 months after stroke. The patients with mRS<3 compared to mRS ≥3 had lower affected hemisphere median DAR (2.5 vs. 7.7), lower affected hemisphere DTABR (1.7 vs. 4.5), lower unaffected hemisphere DAR (2.2 vs. 4.2) and lower unaffected hemisphere DTABR (1.5 vs. 3.2) [16]. The current meta-analysis reveals a higher average effect size estimate for DTABR and DAR for correlation with mRS and NIHSS. DAR and DTABR have similar relationships to clinical outcomes and can be interrelated. We suggest using DTABR or DAR in clinical practice. Cut-offs with high specificity and sensitivity have not been validated in stroke prognostication models. Nevertheless, adding raw DAR or DTABR values to the existing stroke prognostication models, like the motor component of NIHSS and infarct/hematoma volume, may be helpful for prediction. BSI or pdBSI have not been consistently correlated with outcomes and need further validation. There are advantages and disadvantages to the routine use of QEEG in clinical practice for predicting post-stroke recovery. The advantages are the instrument's availability, bedside assessment, and analysis. The disadvantages are an increased burden on the in-house EEG team affecting clinical care, lack of harmonization in the interpretation of analyses, and modest correlation with current assessment tools in small sample studies.

There is variability (1 to 128, Table 1-3) in the number of EEG channels used in the acquisitions; 16 channels were the most common for studies included in the meta-analysis. We did not adjust the meta-analysis results for the number of EEG channels. Depending on the electrode type, dry or gel, the application time increases with the number of channels; thus, in an acute stroke care setting, 8-16 channels may be appropriate to avoid lag in the door-to-reperfusion therapy times. The timing of EEG acquisition for prognostication was also variable (<4.5 h to 10 days of symptom onset, Table 1-3). We did not adjust the meta-analysis results for the timing of EEG acquisition. EEG acquisition for prognostication may be reliable in the first 24-72 h, as observed in the systematic review; however, performing EEG within 24 h post-endovascular thrombectomy may have the ability to assess successful reperfusion compared to futile reperfusion and will account for the effect of the intervention on motor recovery[53].

The studies included in the review, and the meta-analysis has heterogeneity in the EEG assessment of stroke patients [67]. We limited our review to include mRS, NIHSS and FMA as the assessment tools for motor and functional recovery after stroke (the most common functional outcome assessment tools in stroke). These outcome assessment tools, though common, are not without limitations. NIHSS and mRS are global functional deficit and disability assessment scales, respectively. Principles of minimal clinically important difference in these scales may not reflect motor recovery. More studies may consider using upper limb FMA as an assessment tool to describe upper limb motor recovery. There

A: Study or subgroup, Delta-Alpha Ratio (DAR) and Modified Rankin Scale

ID	Effect Size	Lower	Upper	Weight	Weight (%)
Dickey, I	0.27	0.05	0.49	81.08	11.67
Zhang, Pre	0.59	0.28	0.90	40.04	5.76
Zhang, Post	0.51	0.22	0.79	46.89	6.75
Wang	0.03	-0.66	0.72	8.02	1.15
Rogers	0.10	-0.44	0.64	13.22	1.90
Bentes (1), A	0.25	0.13	0.37	276.86	39.83
Bentes (2), A	0.26	0.13	0.39	228.94	32.94
Overall	0.29	0.21	0.36		

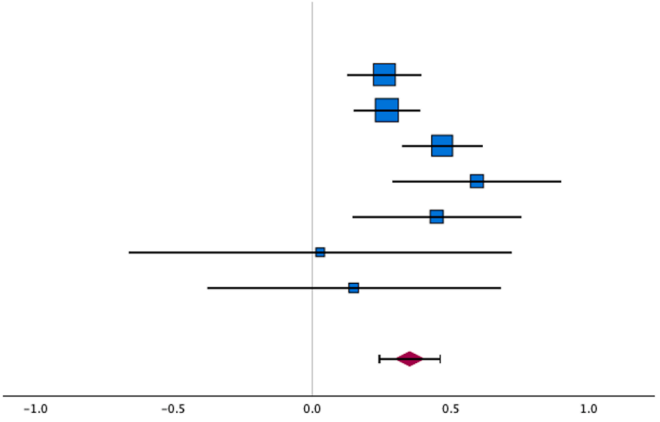


Model: Random-effects model
Homogeneity: Q = 7.62, df = 6, p-value = 0.27
Test of overall effect size: z = 7.55, p-value = 0.00

Footnote: I, Interhemispheric DAR; Pre, EEG acquired pre-endovascular thrombectomy; Post, EEG acquired < 12h post-endovascular thrombectomy; (1), EEG acquired within 72 h of stroke onset; (2), EEG acquired within 7 days of stroke onset; A, affected hemisphere DAR

B: Study or subgroup, Delta-Theta-Alpha-Beta Ratio (DTABR) and Modified Rankin Scale

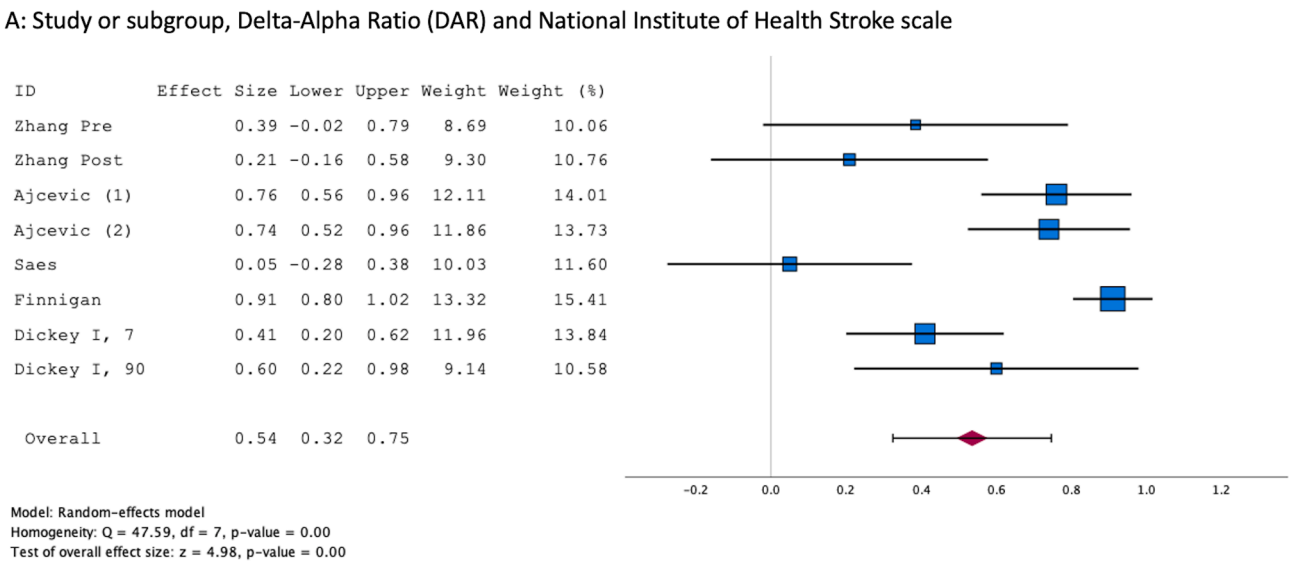
ID	Effect Size	Lower	Upper	Weight	Weight (%)
Bentes (1), A	0.26	0.13	0.39	78.98	24.77
Bentes (2), A	0.27	0.15	0.39	85.04	26.68
Sheorajpanday	0.47	0.32	0.62	73.19	22.96
Zhang, Pre	0.60	0.29	0.90	30.89	9.69
Zhang, Post	0.45	0.14	0.76	30.89	9.69
Wang	0.03	-0.66	0.72	7.54	2.36
Rogers	0.15	-0.38	0.68	12.27	3.85
Overall	0.35	0.24	0.46		



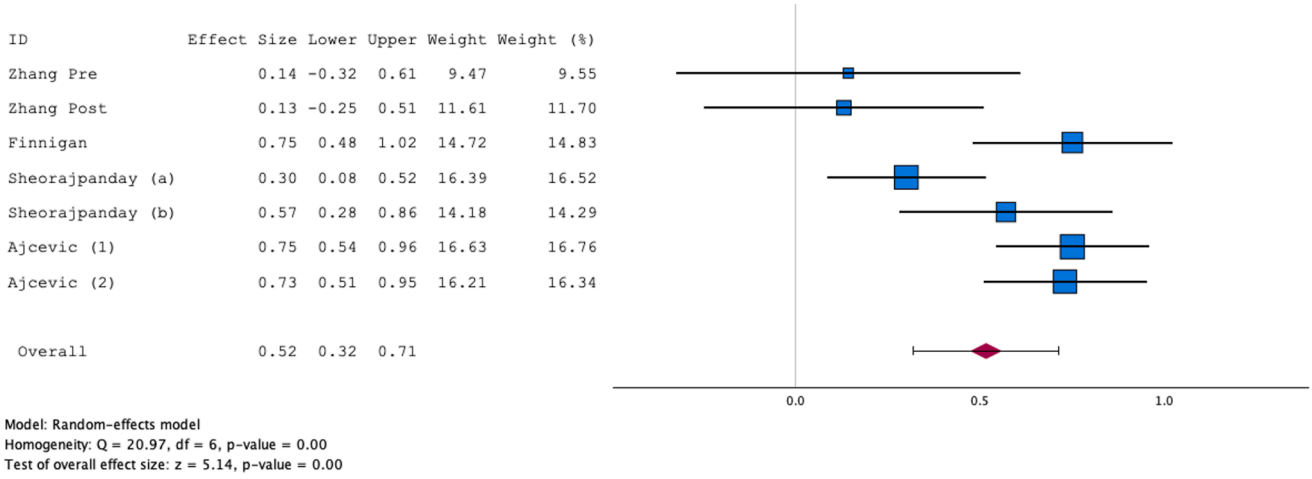
Model: Random-effects model
Homogeneity: Q = 10.12, df = 6, p-value = 0.12
Test of overall effect size: z = 6.29, p-value = 0.00

Footnote: (1), EEG acquired within 72 h of stroke onset; (2), EEG acquired within 7 days of stroke onset; A, affected hemisphere DAR; Pre, EEG acquired pre-endovascular thrombectomy; Post, EEG acquired < 12h post-endovascular thrombectomy;

Fig. 4. Forest plot: A, study or subgroup correlation between baseline Delta-Alpha ratio and follow-up modified Rankin scale; B, study or subgroup correlation between baseline Delta-Alpha ratio and follow-up modified Rankin scale.



Footnote: Pre, EEG acquired pre-endovascular thrombectomy; Post, EEG acquired < 12h post-endovascular thrombectomy; (1), Outcome assessed at 7 days; (2), Outcome assessed at 1 year; I, Interhemispheric DAR; 7, outcome assessed at 7 days; 90, outcome assessed at 90 days



Footnote: Pre, EEG acquired pre-endovascular thrombectomy; Post, EEG acquired < 12h post-endovascular thrombectomy; (a) EEG acquired <24 h of symptom onset; (b) EEG acquired between 24-72 h of symptom onset; (1), Outcome assessed at 7 days; (2), Outcome assessed at 1 year

Fig. 5. Forest plot: A, study or subgroup correlation between baseline Delta-Theta-Alpha-Beta ratio and follow-up National Institute of Health Stroke Scale; B, study or subgroup correlation between baseline Delta-Alpha ratio and follow-up National Institute of Health Stroke Scale.

is a need for harmonized larger sample size multicentre studies with motor recovery-specific scales. Our review focuses on the motor recovery aspect after stroke and does not account for other disabilities that can occur, including sensory deficit, aphasia, neglect, and ataxia. The outcome assessment period is highly variable among the studies included in our review. We assessed all supplementary materials available and reached out to the authors of the selected studies to include all possible indices and their relationship with the selected outcomes.

Conclusion

QEEG indices can give meaningful prognostic information of motor and functional recovery in patients with stroke. QEEG-derived indices DAR and DTABR have the potential to prognosticate post-stroke motor and functional disability. Adding QEEG to the clinical and imaging biomarkers in the acute phase may help better predict post-stroke recovery. A standardized protocol is needed for future studies to increase homogeneity for time and type of EEG assessment method, outcome assessment tool and post-stroke follow-up period for recovery assessment for a meaningful clinical application of QEEG indices. The wider availability of EEG compared to other modalities warrants the creation of new algorithms/decision trees/apps incorporating these indices to predict the prognosis of stroke patients.

CRediT authorship contribution statement

Idha Sood: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Ranjit J Injety:** Writing – review & editing, Validation, Software, Methodology, Formal analysis, Data curation. **Amtul Farheen:** Writing – review & editing, Resources, Methodology. **Setareh Kamali:** Writing – review & editing, Methodology, Data curation. **Ann Jacob:** Writing – review & editing, Methodology, Investigation, Data curation. **Kyle Mathewson:** Writing – review & editing, Validation, Methodology. **Brian H Buck:** Writing –

review & editing, Methodology, Investigation. **Mahesh P Kate:** Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

None.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jstrokecerebrovasdis.2024.108032.

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